

Causal discovery for survey data?

Preliminary data

SHEP-SDOH team meeting

2026-02-26

Agenda and next steps

Risks: ordinal, lower values indicates lower risk.

Needs: binary: no need reported vs unmet/met need

Mental health: ordinal, lower is better

Wellbeing: derived ordinal: four categories: 0-25, 26-51...

- Share polychoric correlations of risks, needs, and outcomes in ONE heatmap.
- Share polychoric correlations of clustered risks, clustered needs, and outcomes in ONE heatmap.
- Graph showing directionality between clustered risks and outcomes.
- Graph showing directionality between clustered needs and outcomes.
- CONSTRAINED graph showing directionality between clustered risks, clustered needs, and outcomes. [??]
- Next steps: simulation study showing need for survey-weighting [??]

Needs with Needs

Needs with risks

Risks with Needs

Risks with risks

Survey-weighted polychoric correlations between needs (n), risks (r), and outcomes (o). n = 5396 complete cases																															
n_HelpBasics	1.00	0.51	0.39	0.68	0.95	0.85	0.75	0.67	0.57	0.54	0.56	0.61	0.61	-0.38	-0.40	0.79	0.45	0.35	0.48	0.80	0.24	0.62	0.32	0.48	0.47	0.41	0.39	0.46	0.44	0.48	0.53
n_HelpCaregive	0.51	1.00	0.37	0.38	0.44	0.48	0.65	0.46	0.45	0.45	0.40	0.50	0.28	-0.32	-0.41	0.40	0.69	0.24	0.17	0.31	0.20	0.49	0.16	0.31	0.32	0.28	0.20	0.36	0.30	0.36	0.22
n_HelpChildcare	0.39	0.37	1.00	0.40	0.49	0.37	0.29	0.45	0.46	0.45	0.48	0.39	0.46	-0.22	-0.23	0.30	0.14	0.84	0.29	0.34	0.11	0.29	0.20	0.34	0.31	0.30	0.46	0.37	0.31	0.30	0.41
n_HelpWork	0.68	0.38	0.40	1.00	0.66	0.62	0.55	0.45	0.55	0.47	0.57	0.61	0.82	-0.36	-0.39	0.57	0.38	0.27	0.83	0.58	0.29	0.43	0.27	0.40	0.45	0.50	0.44	0.45	0.49	0.48	0.68
n_HelpFood	0.95	0.44	0.49	0.66	1.00	0.81	0.73	0.69	0.58	0.54	0.60	0.60	0.65	-0.39	-0.44	0.79	0.40	0.32	0.51	0.87	0.23	0.64	0.35	0.51	0.49	0.41	0.45	0.47	0.45	0.48	0.56
n_HelpHousing	0.85	0.48	0.37	0.62	0.81	1.00	0.74	0.67	0.56	0.54	0.52	0.60	0.56	-0.36	-0.42	0.66	0.47	0.38	0.45	0.63	0.34	0.57	0.39	0.49	0.48	0.39	0.33	0.43	0.47	0.44	0.53
n_HelpTransport	0.75	0.65	0.29	0.55	0.73	0.74	1.00	0.65	0.53	0.50	0.54	0.60	0.52	-0.37	-0.41	0.56	0.53	0.23	0.40	0.56	0.30	0.86	0.30	0.42	0.41	0.40	0.36	0.41	0.34	0.45	0.41
n_HelpInternet	0.67	0.46	0.45	0.45	0.69	0.67	0.65	1.00	0.55	0.50	0.54	0.57	0.47	-0.32	-0.38	0.54	0.40	0.21	0.32	0.59	0.21	0.53	0.55	0.42	0.43	0.38	0.37	0.44	0.38	0.44	0.45
n_HelpIsolation	0.57	0.45	0.46	0.55	0.58	0.56	0.53	0.55	1.00	0.96	0.75	0.48	0.60	-0.68	-0.61	0.43	0.35	0.27	0.48	0.50	0.15	0.45	0.21	0.78	0.79	0.48	0.53	0.48	0.48	0.32	0.53
n_HelpLoneliness	0.54	0.45	0.45	0.47	0.54	0.54	0.50	0.50	0.96	1.00	0.75	0.48	0.63	-0.67	-0.58	0.40	0.39	0.41	0.42	0.46	0.13	0.43	0.19	0.71	0.84	0.46	0.53	0.47	0.47	0.30	0.53
n_HelpDiscrim	0.56	0.40	0.48	0.57	0.60	0.52	0.54	0.54	0.75	0.75	1.00	0.60	0.64	-0.52	-0.51	0.48	0.39	0.28	0.49	0.55	0.13	0.47	0.24	0.59	0.54	0.82	0.74	0.71	0.76	0.46	0.51
n_HelpLegal	0.61	0.50	0.39	0.61	0.60	0.60	0.60	0.57	0.48	0.48	0.60	1.00	0.53	-0.36	-0.36	0.48	0.42	0.23	0.44	0.48	0.24	0.50	0.29	0.37	0.39	0.50	0.36	0.48	0.46	0.92	0.43
n_HelpEduc	0.61	0.28	0.46	0.82	0.65	0.56	0.52	0.47	0.60	0.63	0.64	0.53	1.00	-0.42	-0.44	0.47	0.29	0.25	0.67	0.55	0.18	0.46	0.27	0.44	0.50	0.54	0.52	0.46	0.51	0.42	0.79
o_mh	-0.38	-0.32	-0.22	-0.36	-0.39	-0.36	-0.37	-0.32	-0.68	-0.67	-0.52	-0.36	-0.42	1.00	0.71	-0.36	-0.29	-0.27	-0.38	-0.40	-0.10	-0.34	-0.17	-0.65	-0.70	-0.41	-0.41	-0.37	-0.38	-0.27	-0.41
o_wb	-0.40	-0.41	-0.23	-0.39	-0.44	-0.42	-0.41	-0.38	-0.61	-0.58	-0.51	-0.36	-0.44	0.71	1.00	-0.40	-0.35	-0.21	-0.46	-0.48	-0.16	-0.40	-0.24	-0.66	-0.65	-0.40	-0.42	-0.39	-0.37	-0.29	-0.46
r_PayBasics	0.79	0.40	0.30	0.57	0.79	0.66	0.56	0.54	0.43	0.40	0.48	0.48	0.47	-0.36	-0.40	1.00	0.51	0.36	0.49	0.85	0.20	0.56	0.34	0.46	0.43	0.40	0.36	0.45	0.41	0.41	0.51
r_GetCaregiving	0.45	0.69	0.14	0.38	0.40	0.47	0.53	0.40	0.35	0.39	0.39	0.42	0.29	-0.29	-0.35	0.51	1.00	0.45	0.20	0.41	0.17	0.53	0.23	0.41	0.35	0.31	0.23	0.39	0.31	0.30	0.30
r_GetChildcare	0.35	0.24	0.84	0.27	0.32	0.38	0.23	0.21	0.27	0.41	0.28	0.23	0.25	-0.27	-0.21	0.36	0.45	1.00	0.21	0.25	0.11	0.29	0.22	0.36	0.33	0.18	0.29	0.27	0.28	0.13	0.35
r_FindWork	0.48	0.17	0.29	0.83	0.51	0.45	0.40	0.32	0.48	0.42	0.49	0.44	0.67	-0.38	-0.46	0.49	0.20	0.21	1.00	0.55	0.19	0.33	0.22	0.46	0.49	0.44	0.47	0.37	0.44	0.35	0.66
r_LackFood	0.80	0.31	0.34	0.58	0.87	0.63	0.56	0.59	0.50	0.46	0.55	0.48	0.55	-0.40	-0.48	0.85	0.41	0.25	0.55	1.00	0.20	0.59	0.35	0.56	0.52	0.46	0.45	0.45	0.47	0.44	0.57
r_HaveHousing	0.24	0.20	0.11	0.29	0.23	0.34	0.30	0.21	0.15	0.13	0.13	0.24	0.18	-0.10	-0.16	0.20	0.17	0.11	0.19	0.20	1.00	0.27	0.48	0.03	0.16	0.16	0.16	0.14	0.10	0.20	0.04
r_LackTransport	0.62	0.49	0.29	0.43	0.64	0.57	0.86	0.53	0.45	0.43	0.47	0.50	0.46	-0.34	-0.40	0.56	0.53	0.29	0.33	0.59	0.27	1.00	0.34	0.44	0.39	0.39	0.38	0.42	0.36	0.42	0.40
r_AccessInternet	0.32	0.16	0.20	0.27	0.35	0.39	0.30	0.55	0.21	0.19	0.24	0.29	0.27	-0.17	-0.24	0.34	0.23	0.22	0.22	0.35	0.48	0.34	1.00	0.15	0.20	0.20	0.22	0.25	0.25	0.21	0.32
r_LackSocial	0.48	0.31	0.34	0.40	0.51	0.49	0.42	0.42	0.78	0.71	0.59	0.37	0.44	-0.65	-0.66	0.46	0.41	0.36	0.46	0.56	0.03	0.44	0.15	1.00	0.83	0.45	0.47	0.48	0.50	0.28	0.50
r_FeelLonely	0.47	0.32	0.31	0.45	0.49	0.48	0.41	0.43	0.79	0.84	0.54	0.39	0.50	-0.70	-0.65	0.43	0.35	0.33	0.49	0.52	0.16	0.39	0.20	0.83	1.00	0.42	0.45	0.43	0.45	0.32	0.50
r_DiscrimREC	0.41	0.28	0.30	0.50	0.41	0.39	0.40	0.38	0.48	0.46	0.82	0.50	0.54	-0.41	-0.40	0.40	0.31	0.18	0.44	0.46	0.16	0.39	0.20	0.45	0.42	1.00	0.78	0.68	0.72	0.43	0.46
r_DiscrimGend	0.39	0.20	0.46	0.44	0.45	0.33	0.36	0.37	0.53	0.53	0.74	0.36	0.52	-0.41	-0.42	0.36	0.23	0.29	0.47	0.45	0.16	0.38	0.22	0.47	0.45	0.78	1.00	0.69	0.69	0.34	0.45
r_DiscrimAge	0.46	0.36	0.37	0.45	0.47	0.43	0.41	0.44	0.48	0.47	0.71	0.48	0.46	-0.37	-0.39	0.45	0.39	0.27	0.37	0.45	0.14	0.42	0.25	0.48	0.43	0.68	0.69	1.00	0.74	0.38	0.48
r_DiscrimOther	0.44	0.30	0.31	0.49	0.45	0.47	0.34	0.38	0.48	0.47	0.76	0.46	0.51	-0.38	-0.37	0.41	0.31	0.28	0.44	0.47	0.10	0.36	0.25	0.50	0.45	0.72	0.69	0.74	1.00	0.33	0.49
r_LegalNeed	0.48	0.36	0.30	0.48	0.48	0.44	0.45	0.44	0.32	0.30	0.46	0.92	0.42	-0.27	-0.29	0.41	0.30	0.13	0.35	0.44	0.20	0.42	0.21	0.28	0.32	0.43	0.34	0.38	0.33	1.00	0.38
r_LackEduc	0.53	0.22	0.41	0.68	0.56	0.53	0.41	0.45	0.53	0.53	0.51	0.43	0.79	-0.41	-0.46	0.51	0.30	0.35	0.66	0.57	0.04	0.40	0.32	0.50	0.50	0.46	0.45	0.48	0.49	0.38	1.00
n_HelpBasics	n_HelpCaregive	n_HelpChildcare	n_HelpWork	n_HelpFood	n_HelpHousing	n_HelpTransport	n_HelpInternet	n_HelpIsolation	n_HelpLoneliness	n_HelpDiscrim	n_HelpLegal	n_HelpEduc	o_mh	o_wb	r_PayBasics	r_GetCaregiving	r_GetChildcare	r_FindWork	r_LackFood	r_HaveHousing	r_LackTransport	r_AccessInternet	r_LackSocial	r_FeelLonely	r_DiscrimREC	r_DiscrimGend	r_DiscrimAge	r_DiscrimOther	r_LegalNeed	r_LackEduc	

Clustering information

FOR NEEDS

n_cluster_basics: n_housing, n_basics, and n_food

n_cluster_work_edu: n_work and n_education

n_clust_discrim_iso: n_discrimination, n_isolation, n_loneliness

n_childcare, n_caregiving, n_legal, n_transport, and n_internet treated independently

FOR RISKS

r_cluster_basics: r_basics, and r_food

r_cluster_work_edu: r_work and r_education

r_clust_iso: r_isolation, r_loneliness

r_clust_discrimination: four r_discriminations

r_childcare, r_caregiving, and r_legal, r_transport and r_internet treated independently

Survey-weighted polychoric correlations between clustered needs (n), clustered risks (r), and outcomes (o).
n = 5396 complete cases

Needs with Needs

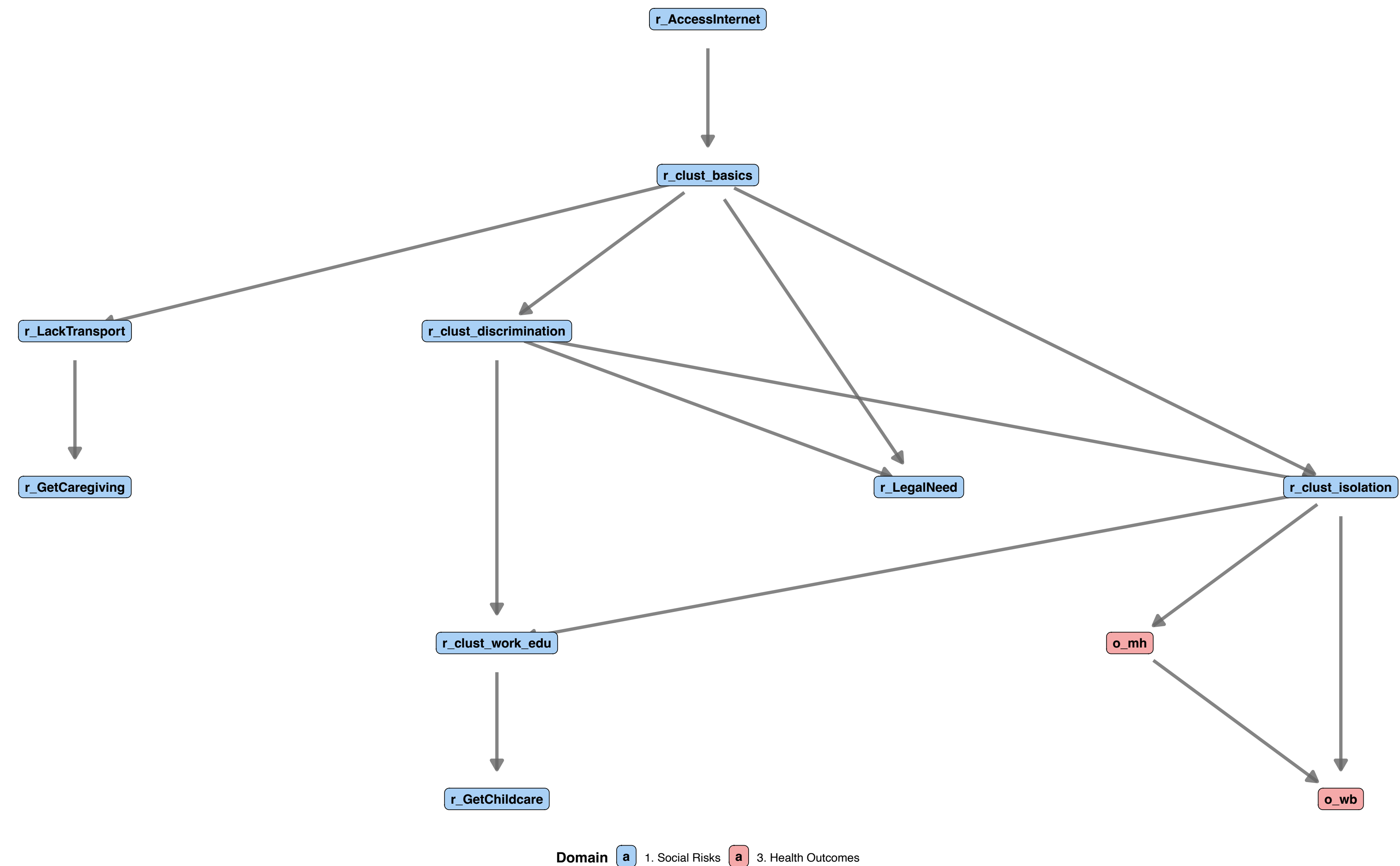
Needs with risks

Risks with Needs

Risks with risks

n_clust_basics	1.00	0.57	0.67	0.44	0.46	0.69	0.60	0.75	−0.38	−0.43	0.37	0.77	0.47	0.53	0.60	0.42	0.39	0.64	0.48
n_clust_discrim_iso	0.57	1.00	0.60	0.47	0.45	0.52	0.50	0.50	−0.69	−0.61	0.19	0.39	0.58	0.83	0.51	0.36	0.29	0.45	0.34
n_clust_work_edu	0.67	0.60	1.00	0.31	0.49	0.48	0.57	0.57	−0.41	−0.43	0.27	0.53	0.58	0.49	0.83	0.34	0.28	0.48	0.44
n_HelpCaregive	0.44	0.47	0.31	1.00	0.37	0.46	0.50	0.65	−0.32	−0.41	0.16	0.36	0.35	0.32	0.24	0.69	0.24	0.49	0.36
n_HelpChildcare	0.46	0.45	0.49	0.37	1.00	0.45	0.39	0.29	−0.22	−0.23	0.20	0.26	0.38	0.34	0.40	0.14	0.84	0.29	0.30
n_HelpInternet	0.69	0.52	0.48	0.46	0.45	1.00	0.57	0.65	−0.32	−0.38	0.55	0.53	0.44	0.43	0.41	0.40	0.21	0.53	0.44
n_HelpLegal	0.60	0.50	0.57	0.50	0.39	0.57	1.00	0.60	−0.36	−0.36	0.29	0.48	0.55	0.40	0.45	0.42	0.23	0.50	0.92
n_HelpTransport	0.75	0.50	0.57	0.65	0.29	0.65	0.60	1.00	−0.37	−0.41	0.30	0.55	0.42	0.43	0.43	0.53	0.23	0.86	0.45
o_mh	−0.38	−0.69	−0.41	−0.32	−0.22	−0.32	−0.36	−0.37	1.00	0.71	−0.17	−0.32	−0.45	−0.72	−0.41	−0.29	−0.27	−0.34	−0.27
o_wb	−0.43	−0.61	−0.43	−0.41	−0.23	−0.38	−0.36	−0.41	0.71	1.00	−0.24	−0.41	−0.45	−0.67	−0.46	−0.35	−0.21	−0.40	−0.29
r_AccessInternet	0.37	0.19	0.27	0.16	0.20	0.55	0.29	0.30	−0.17	−0.24	1.00	0.47	0.28	0.20	0.28	0.23	0.22	0.34	0.21
r_clust_basics	0.77	0.39	0.53	0.36	0.26	0.53	0.48	0.55	−0.32	−0.41	0.47	1.00	0.44	0.41	0.49	0.40	0.18	0.57	0.41
r_clust_discrimination	0.47	0.58	0.58	0.35	0.38	0.44	0.55	0.42	−0.45	−0.45	0.28	0.44	1.00	0.52	0.56	0.37	0.30	0.46	0.46
r_clust_isolation	0.53	0.83	0.49	0.32	0.34	0.43	0.40	0.43	−0.72	−0.67	0.20	0.41	0.52	1.00	0.55	0.40	0.38	0.44	0.31
r_clust_work_edu	0.60	0.51	0.83	0.24	0.40	0.41	0.45	0.43	−0.41	−0.46	0.28	0.49	0.56	0.55	1.00	0.29	0.34	0.39	0.37
r_GetCaregiving	0.42	0.36	0.34	0.69	0.14	0.40	0.42	0.53	−0.29	−0.35	0.23	0.40	0.37	0.40	0.29	1.00	0.45	0.53	0.30
r_GetChildcare	0.39	0.29	0.28	0.24	0.84	0.21	0.23	0.23	−0.27	−0.21	0.22	0.18	0.30	0.38	0.34	0.45	1.00	0.29	0.13
r_LackTransport	0.64	0.45	0.48	0.49	0.29	0.53	0.50	0.86	−0.34	−0.40	0.34	0.57	0.46	0.44	0.39	0.53	0.29	1.00	0.42
r_LegalNeed	0.48	0.34	0.44	0.36	0.30	0.44	0.92	0.45	−0.27	−0.29	0.21	0.41	0.46	0.31	0.37	0.30	0.13	0.42	1.00
	n_clust_basics	n_clust_discrim_iso	n_clust_work_edu	n_HelpCaregive	n_HelpChildcare	n_HelpInternet	n_HelpLegal	n_HelpTransport	o_mh	o_wb	r_AccessInternet	r_clust_basics	r_clust_discrimination	r_clust_isolation	r_clust_work_edu	r_GetCaregiving	r_GetChildcare	r_LackTransport	r_LegalNeed

DAG 1 (Risks → Outcomes)



Out of all the risks, the algorithm identified Social Isolation/Loneliness (**r_clust_isolation**) as the sole, direct upstream driver of Mental Health (**o_mh**).

Risks like lacking internet (**r_AccessInternet**) and basic needs (**r_clust_basics**) trigger downstream social fractures, specifically discrimination (**r_clust_discrimination**), which in turn directly causes severe social isolation (**r_clust_isolation**), ultimately damaging mental health.

General wellbeing (**o_wb**) is shown as a final terminal node, driven simultaneously by a person's mental health state (**o_mh**) and the direct burden of their social isolation.

Tree structure, arrows point downwards

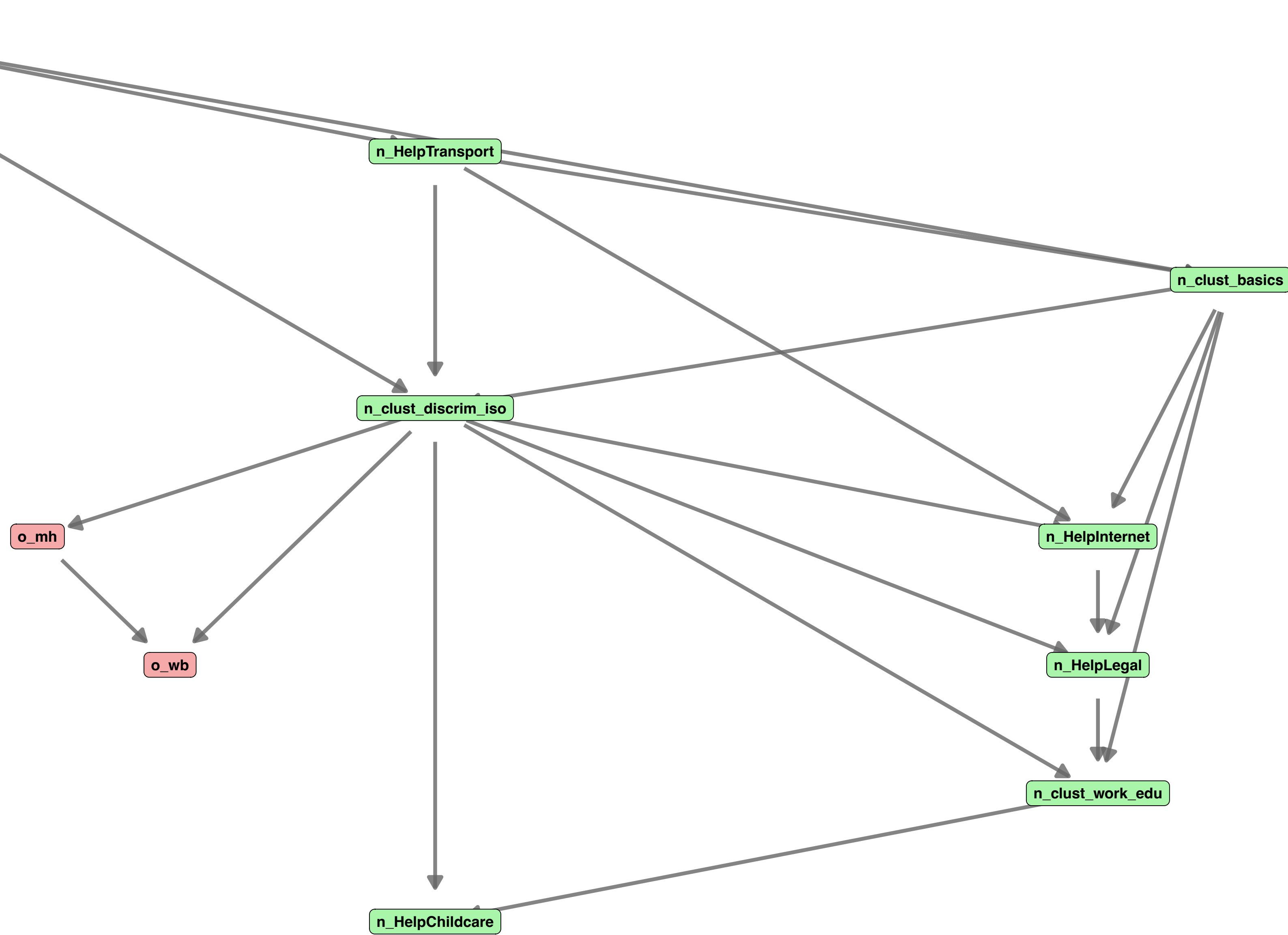
Will make cleaner plot later

DAG 2 (Needs -> Outcomes)

n_clust_discrim_iso (the expressed need for help with isolation and discrimination) emerges as a central hub. It is the end-result of a chain of physical needs (caregiving → transport → basics), indicating that physical unmet needs quickly compound into isolation/discrimination induced stress.

Just as in DAG 1, this isolation (and discrimination domain (**n_clust_discrim_iso**) acts as the exclusive direct trigger for adverse Mental Health (**o_mh**) and Wellbeing (**o_wb**) outcomes.

Just as in DAG 1, wellbeing (**o_wb**) is shown as a final terminal node, driven simultaneously by a person's mental health state (**o_mh**) and the direct burden of their social isolation.

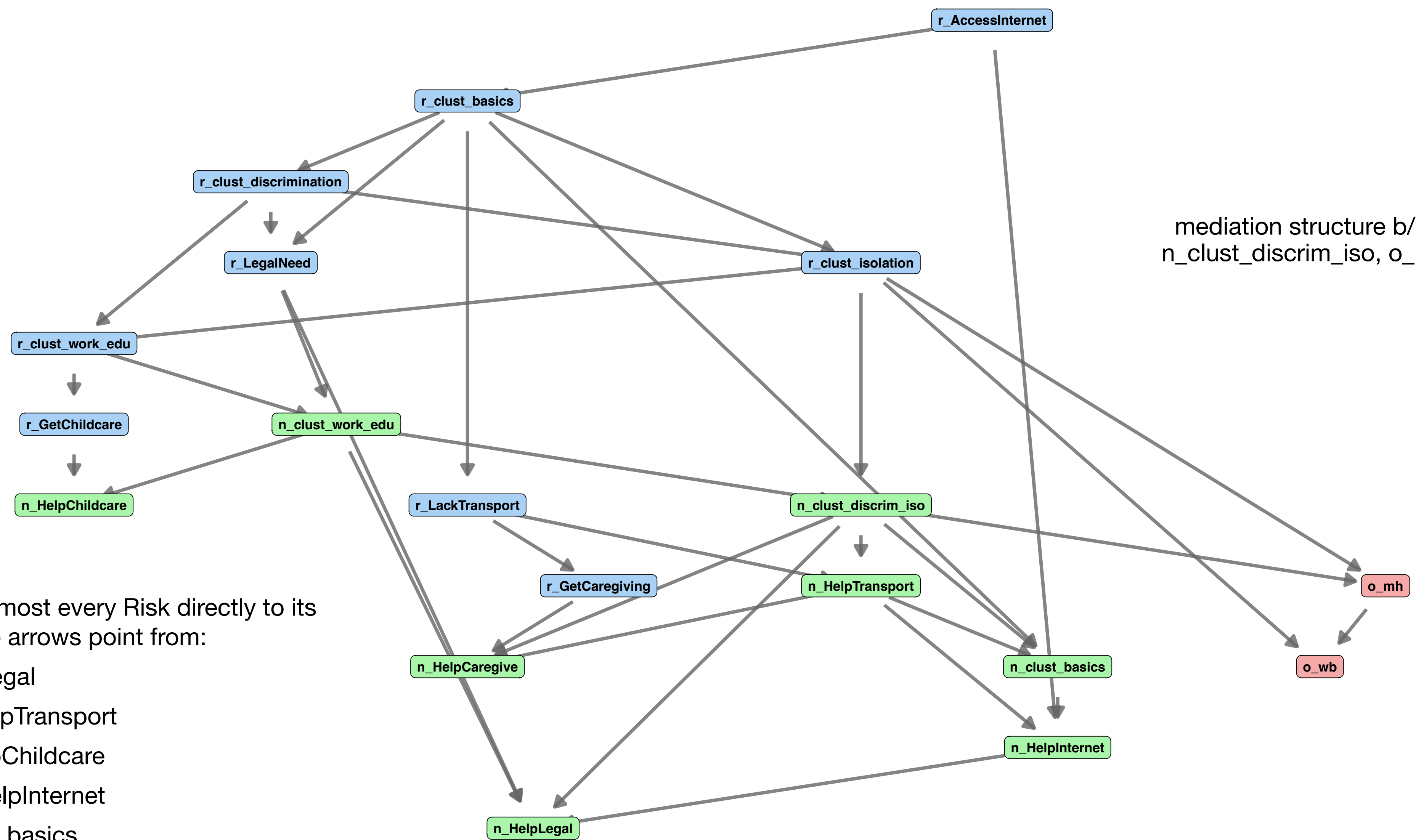


Domain a 2. Social Needs a 3. Health Outcomes

Will make cleaner plot later

Tree structure, arrows point downwards

DAG 3 (Risks -> Needs -> Outcomes)



mediation structure b/w r_clust_iso, n_clust_discrim_iso, o_mh, and o_wb.

The algorithm mapped almost every Risk directly to its corresponding Need. The arrows point from:

- r_LegalNeed → n_HelpLegal
- r_LackTransport → n_HelpTransport
- r_GetChildcare → n_HelpChildcare
- r_AccessInternet → n_HelpInternet
- r_clust_basics → n_clust_basics
- r_clust_work_edu → n_clust_work_edu

SHEP survey accurately captures a progression from an underlying risk state to an active need state within specific social domains.

Domain a 1. Social Risks a 2. Social Needs a 3. Health Outcomes

Will make cleaner plot later
Tree structure, arrows point downwards

Agenda and next steps

- Share polychoric correlations of risks, needs, and outcomes in ONE heatmap.
- Share polychoric correlations of clustered risks, clustered needs, and outcomes in ONE heatmap.

These findings are based on one algorithm which is good for ordinal data + with survey weighting. Should we compare some competing models/algorithms for a paper? Looking for ideas on how to position: only novelty so far is just survey-weighting. Happy to do more analyses as well!

- CONSTRAINED graph showing directionality between clustered risks, clustered needs, and outcomes. [??]
- **Next steps: simulation study showing need for survey-weighting.**